Big Data Analytics

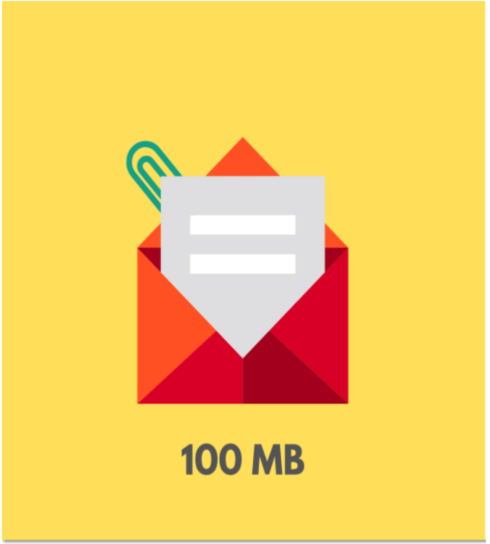
What is big data?

Big Data refers to a huge volume of data, that cannot be stored and processed using the traditional computing approach within a given time frame.

Example of big data?

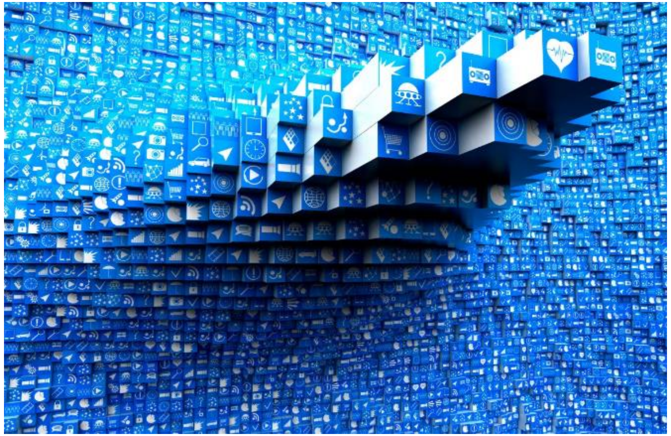
For example, if we try to attach a document that is of 100 megabytes in size to an email we would not be able to do so. As the email system would not support an attachment of this size.

Therefore this 100 megabytes of attachment with respect to email can be referred to as Big Data.



What is big data Analytics?

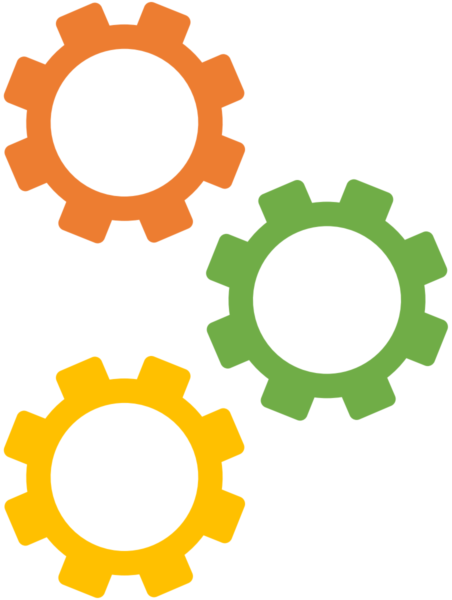
Big data Analytics is a process to extract meaningful insight from big such as hidden patterns, unknown correlations, market trends and customer preferences



Types of Big Data Analytics/ Digital Data

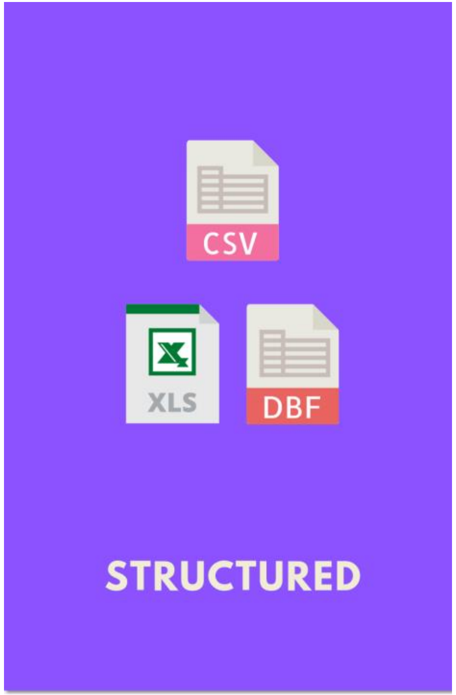
Structured Data

Unstructured Data



Semi-Structured Data

Structured Data



Structured Data refers to the data that has a proper structure associated with it. For example, the data that is present within the databases, the CSV files, and the excel spreadsheets can be referred to as Structured Data.

Employee_ID	Employee_Name	Gender	Department	Salary_In_lacs
2365	Rajesh Kulkarni	Male	Finance	650000
3398	Pratibha Joshi	Female	Admin	650000
7465	Shushil Roy	Male	Admin	500000
7500	Shubhojit Das	Male	Finance	500000
7699	Priya Sane	Female	Finance	550000

Easy to Work with Structured Data

Structured data is easier to work with because of the following reasons:

1. **Easy Data Operations:**
   You can easily insert, update, or delete data using simple commands (DML - Data Manipulation Language). This helps in storing, accessing, and analyzing data efficiently.
2. **Security:**
   Structured data can be protected using strong encryption and security measures. Only authorized users can access or view sensitive information.

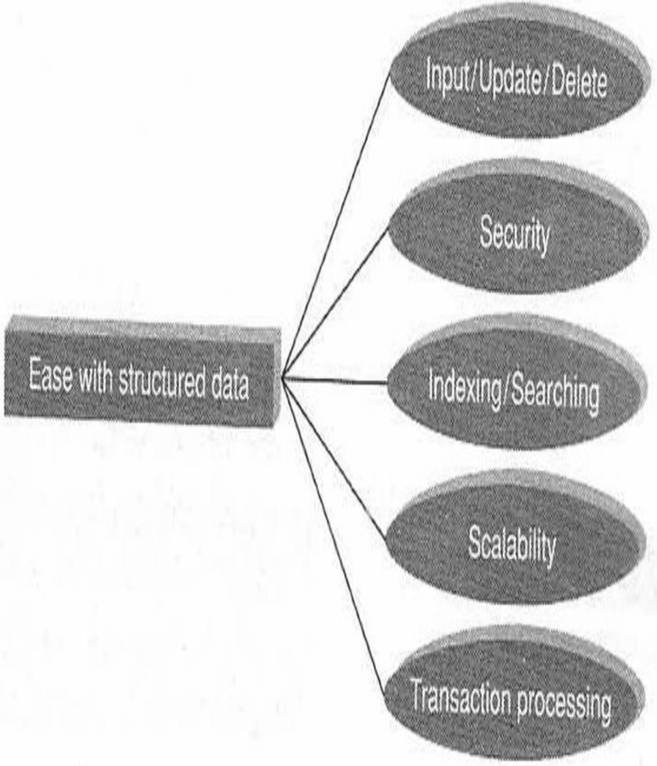


Figure 1.5 Ease of working with structured data.

3. Faster Searching with Indexing:
Indexing helps in finding data quickly. Although it may use more space, it improves search performance.
4. Scalability:
You can easily increase storage and processing power in structured databases by upgrading the server or hardware.
5. Reliable Transactions (ACID Properties):
Structured data systems follow ACID rules to make transactions safe and reliable:

1. **Atomicity:** A task either completes fully or not at all.
2. **Consistency:** Data remains accurate before and after a transaction.
3. **Isolation:** Each transaction runs independently.
4. **Durability:** Once data is saved, it stays safe even after a system crash.

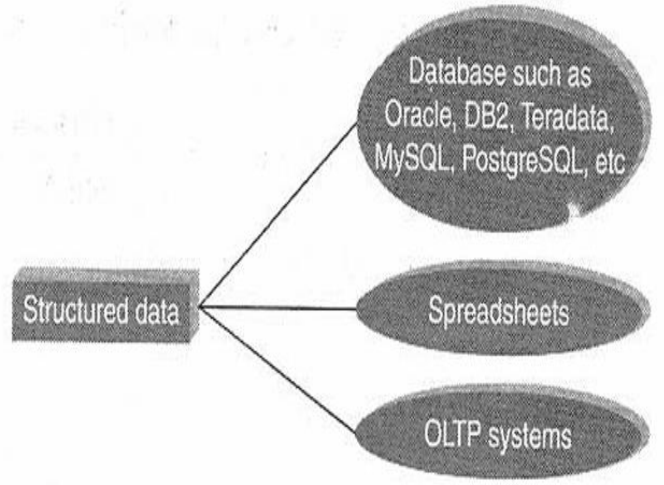
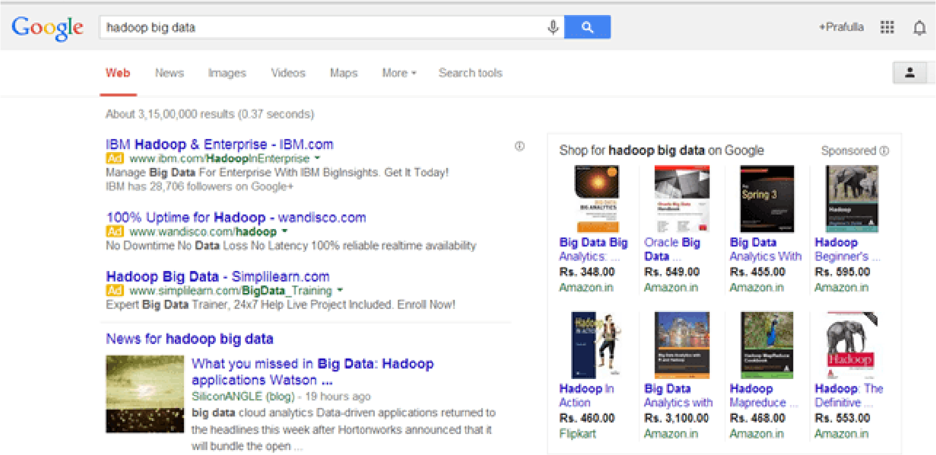


Figure 1.4 Sources of structured data.

Unstructured Data

Un-Structured Data refers to the data that does not have any structure associated with it at all. For example, the image files, the audio files, and the video files can be referred to as Un-Structured Data.



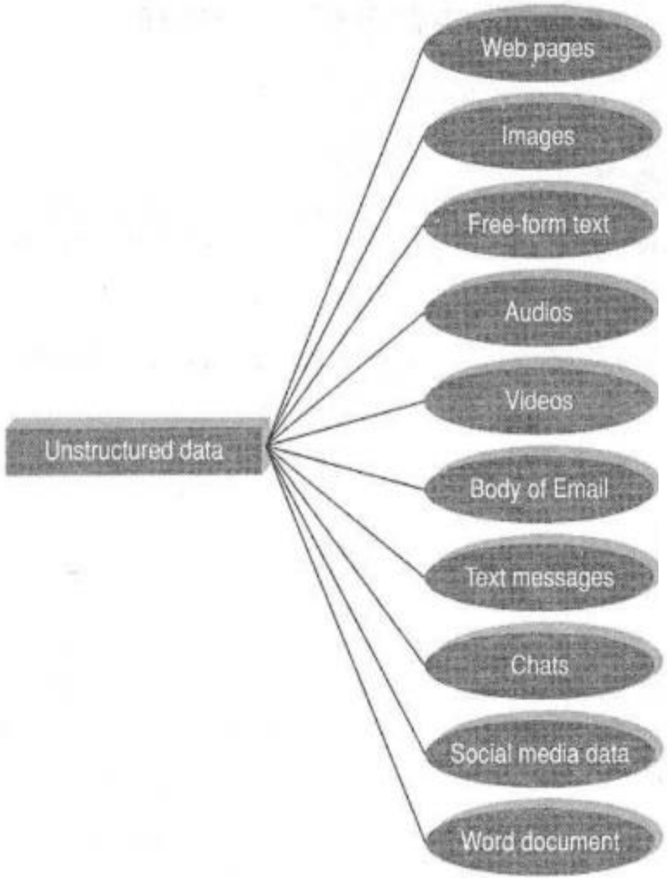


Figure 1.8 Sources of unstructured data.

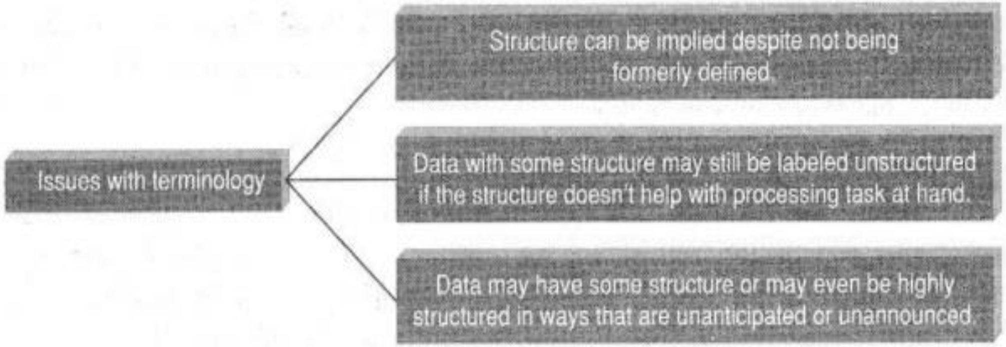


Figure 1.9 Issues with terminology of unstructured data.

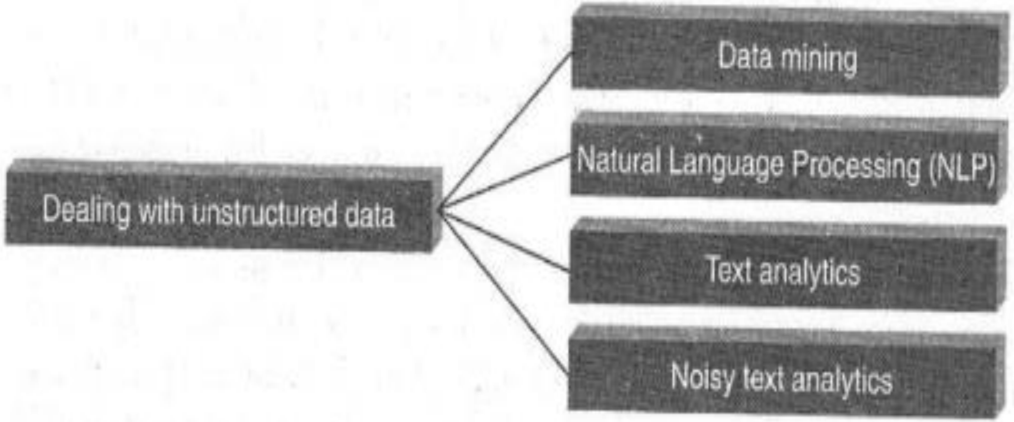


Figure 1.11 Dealing with unstructured data.

Techniques to Find Patterns in Unstructured Data

1. Data Mining:

- Involves analyzing large datasets.
- Uses AI, machine learning, statistics, and database tools to find patterns or relationships.
- Part of the process called "knowledge discovery in databases.“

2. Popular Data Mining Techniques:

•Association Rule Mining:

Also called market basket or affinity analysis. It finds which items are bought together.

Example: If someone buys bread, they might also buy eggs or cheese.

•Regression Analysis:

Predicts relationships between two variables.

One is the dependent variable (to be predicted), and others are independent variables (used for prediction).

•Collaborative Filtering:

Predicts user preferences based on similar users' choices.

Example: If User 4 likes videos and is similar to others who like videos, it's likely User 4 will also prefer videos.

2. Text Analytics or Text Mining:

- Deals with unstructured text data, which is hard to handle.
- Finds useful and meaningful patterns in text.
- Tasks include: categorizing text, clustering, sentiment analysis, and identifying concepts/entities.

3. Natural Language Processing (NLP):

Helps computers understand and process human language.

4. Noisy Text Analytics:

- Extracts information from messy or unstructured text like chats, emails, or messages.

- Deals with spelling errors, short forms, missing punctuation, and filler words like "uh", "um", etc.

5. Manual Tagging with Metadata:

Manually adds labels or tags to unstructured data to help understand it.

6. Part-of-Speech (POS) Tagging:

Identifies the grammatical role of each word (like noun, verb, adjective) in a sentence.

7. Unstructured Information Management Architecture (UIMA):

- An open-source tool from IBM.
- Used for analyzing and understanding text in real-time.
- Helps find meaning and relationships in unstructured data.

Semi-structured Data

Semi-Structured Data refers to the data that does not have a proper structure associated with it. For example, the data that is present within the emails, the log files, and the word documents can be referred to as Semi-Structured Data.

```
<rec><name>Prashant Rao</name><sex>Male</sex><age>35</age></rec>
<rec><name>Seema R.</name><sex>Female</sex><age>41</age></rec>
<rec><name>Satish Mane</name><sex>Male</sex><age>29</age></rec>
<rec><name>Subrato Roy</name><sex>Male</sex><age>26</age></rec>
<rec><name>Jeremiah J.</name><sex>Male</sex><age>35</age></rec>
```

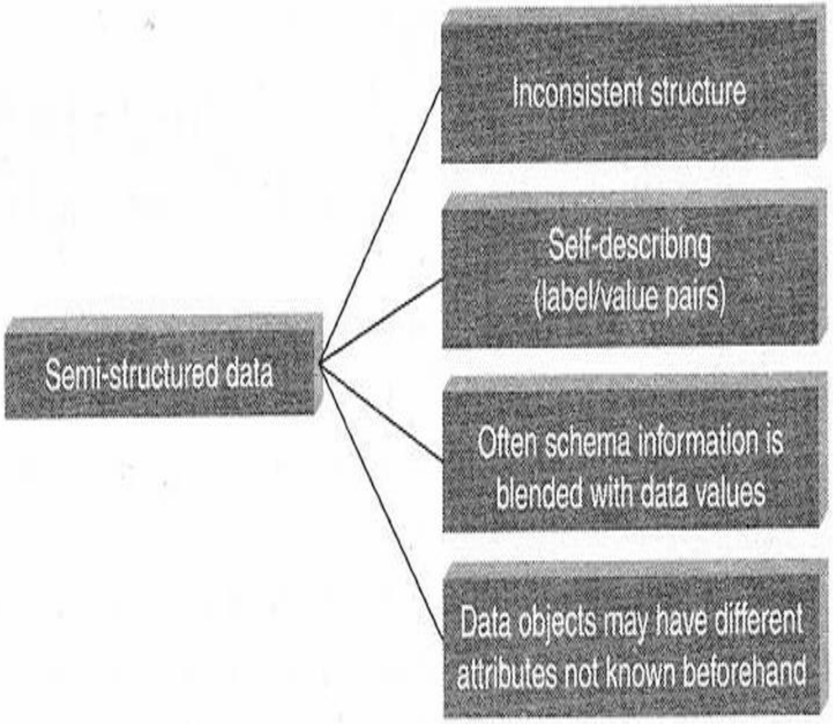


Figure 1.6 Characteristics of semi-structured data.

Sources of Semi-Structured Data
The two main types of semi-structured data formats are **XML** and **JSON**.
1.XML (eXtensible Markup Language): XML is often used in web services and follows SOAP (Simple Object Access Protocol) rules.
2.JSON (JavaScript Object Notation): JSON is used to send data between a server and a website. It is popular with REST-based web services. Databases like MongoDB and Couchbase use JSON to store data.

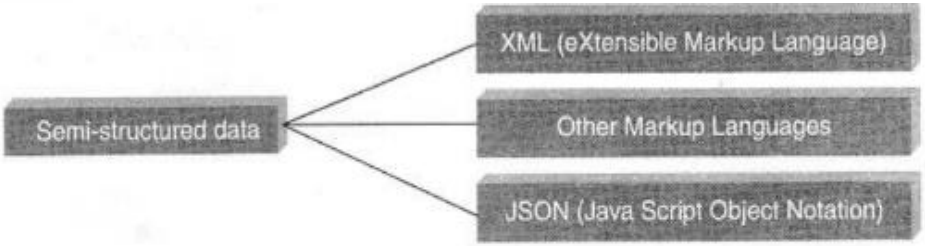


Figure 1.7 Sources of semi-structured data.

```json
{
  "OrderID": "A12345",
  "Customer": {
    "Name": "John Doe",
    "Phone": "+123456789",
    "Address": "123 Main Street, City"
  },
  "Items": [
    {"ProductID": "P1001", "ProductName": "Smartphone", "Quantity": 1},
    {"ProductID": "P2001", "ProductName": "Phone Case", "Quantity": 2}
  ],
  "ShipmentStatus": [
    {"Status": "Packed", "Timestamp": "2025-08-01T10:00:00Z"},
    {"Status": "Shipped", "Timestamp": "2025-08-02T15:30:00Z"}
  ],
  "GiftMessage": "Happy Birthday!"
}
```

xml

```xml
<Order>
  <OrderID>1001</OrderID>
  <CustomerName>Alice</CustomerName>
  <GiftMessage>Happy Birthday!</GiftMessage>  <!-- This tag is optional -->
</Order>

<Order>
  <OrderID>1002</OrderID>
  <CustomerName>Bob</CustomerName>
  <!-- This order has no GiftMessage tag -->
</Order>
```

- It does not conform to the data models that one typically associates with relational databases or any other form of data tables.
- It uses tags to segregate **semantic elements**.
- Tags are also used **to enforce hierarchies of records and fields within data**.
- There is no separation between the data and the schema.
- The amount of structure used is dictated by the purpose at hand.
- In semi-structured data, **entities belonging to the same class and also grouped together need not necessarily have the same set of attributes**.
- And if at all, they have the same set of attributes, the order of attributes may not be similar and for all practical purposes it is not important as well.

Characteristics of Big Data

Big Data is categorized into 3 important characteristics.

- Volume
- Velocity
- Variety

Evolution of Big Data

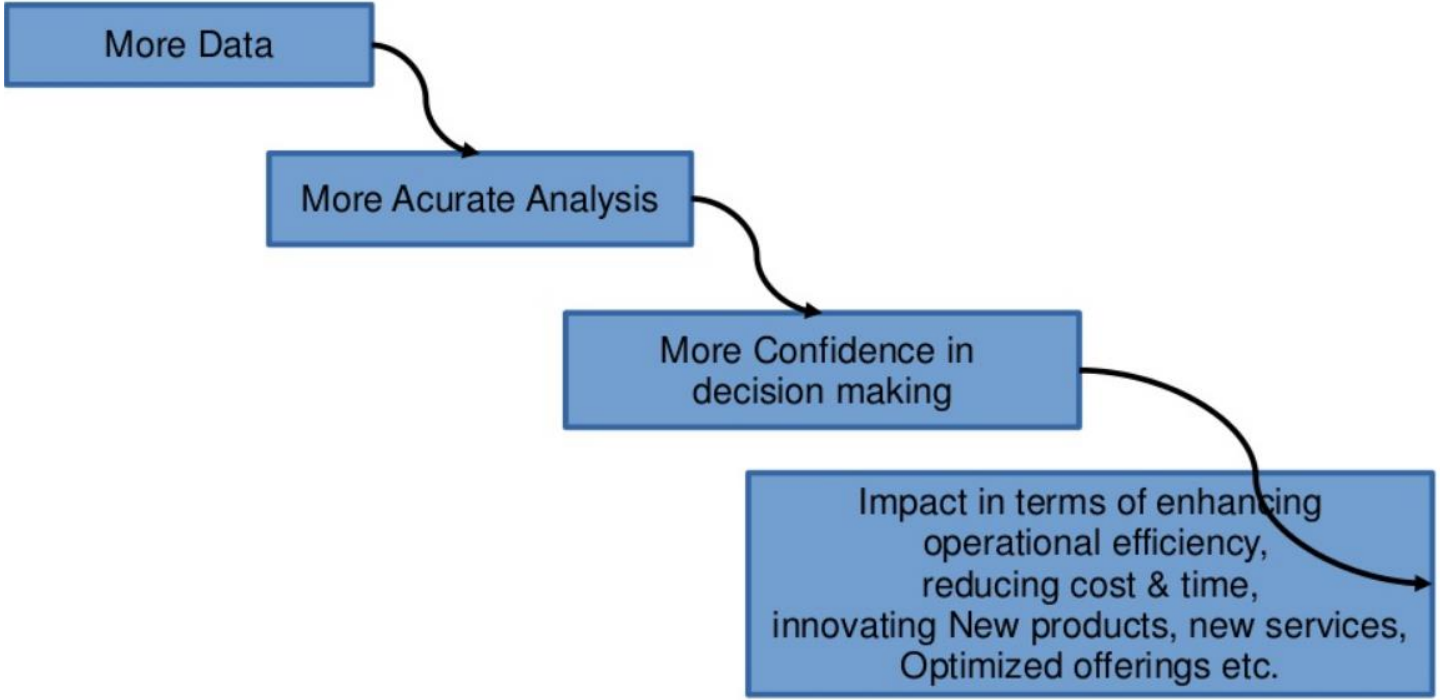
Table 2.1 The evolution of big data

	Data Generation and Storage	Data Utilization	Data Driven
Complex and Unstructured			Structured data, unstructured data, multimedia data
Complex and Relational		Relational databases: Data-intensive applications	
Primitive and Structured	Mainframes: Basic data storage		
	1970s and before	Relational (1980s and 1990s)	2000s and beyond

Other characteristics of data which are not definitional for Big Data

- Veracity and Validity : deals with abnormality, accuracy and correctness
- Volatility : deals with data validity
- Variability : deals with data flow which is highly inconsistent

Why Big Data?


More Data
More Acurate Analysis
More Confidence in decision making
Impact in terms of enhancing operational efficiency, reducing cost & time, innovating New products, new services, Optimized offerings etc.

Chracteristics of Data

- Composition: Deals with the structure of data. i.e.
  - sources of data, types of data, nature of data.
  - Granularity (Quality of data)
  - Types & nature of data (Static or Real Time)

- Condition: Deals with state of data i.e.
  - Can one use this data as it is for analysis
  - (or) does it require cleaning for further enhancemment and enr

- Context: deals with generation of data, sensitivity of data. i.e
  - Where has this data been generated?
  - Why was this data generated?
  - How sensitive is this data (health data, financial stmt)?
  - What are the events associated with this data ? And so on...

Definition of big data

- ***"big data is high-volume, high-velocity, and high-variety information assets."*** This refers to large-scale data (massive volumes) that comes in diverse forms—a mix of structured, semi-structured, and unstructured formats. Such data demands efficient speed and capacity for storage, preparation, processing, and analysis.
- ***"cost-effective, innovative forms of information processing."*** It emphasizes adopting new methods and technologies to collect (ingest), store, process, persist, integrate, and visualize data characterized by high volume, high velocity, and high variety.
- ***"enhanced insight and decision making."*** This focuses on extracting deeper, richer, and more meaningful insights, and then applying them to make quicker and more accurate decisions—ultimately creating business value and gaining a competitive advantage.

Data → Information → Actionable Intelligence → Better Decisions → Enhanced Business Value

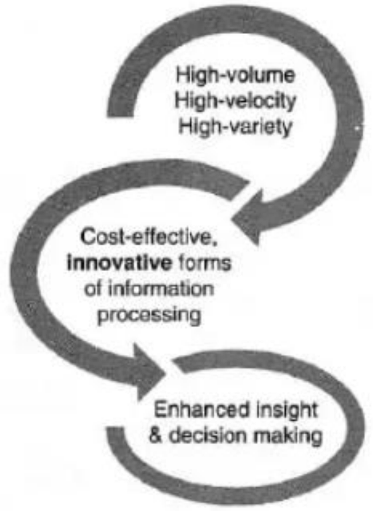


Figure 2.3 Definition of big data – Gartner.

Volume

Volume refers to the amount of data that is getting generated.

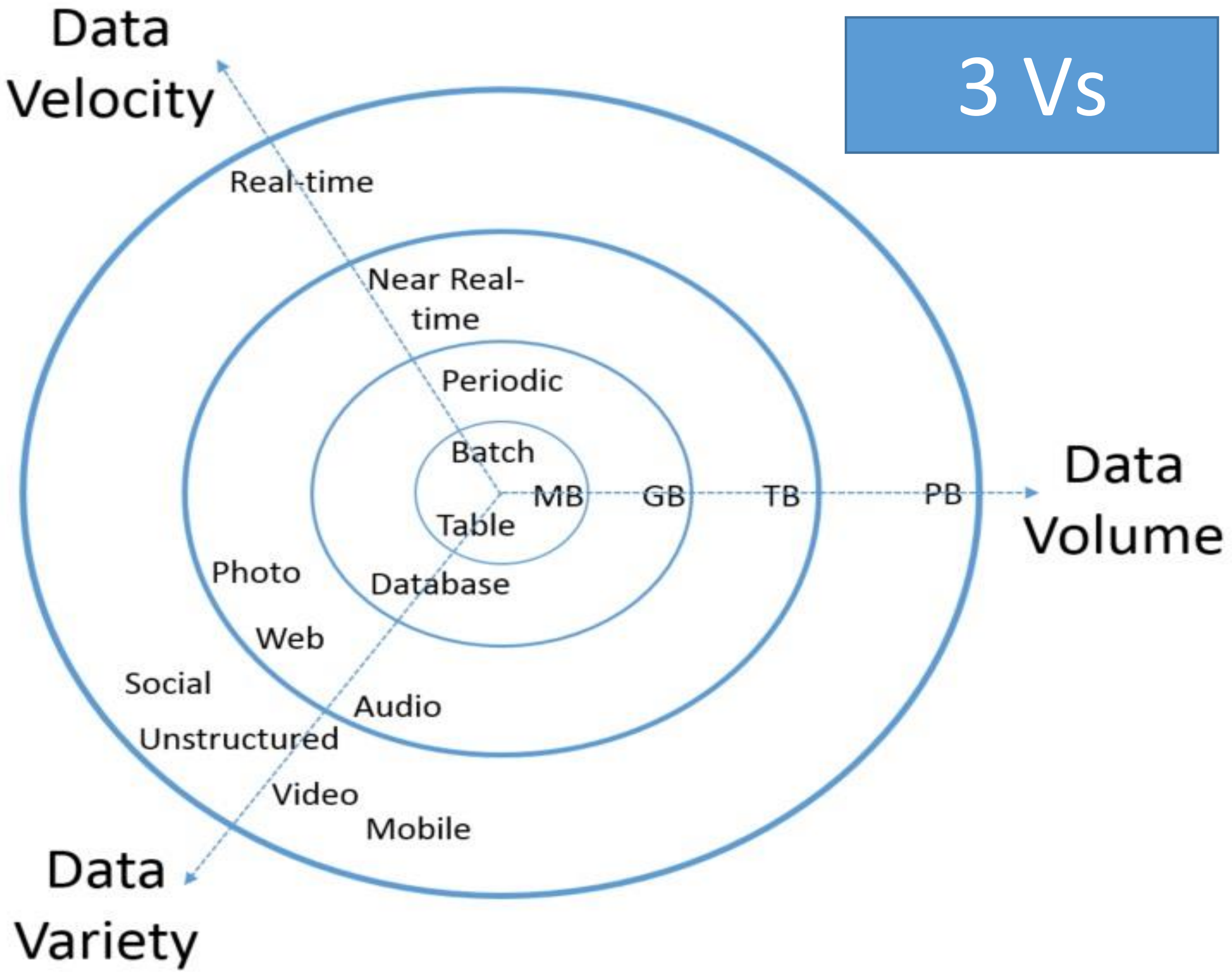
Velocity

Velocity refers to the speed at which the data is getting generated.

Variety

Variety refers to the different types of data that is getting generated.

3 Vs



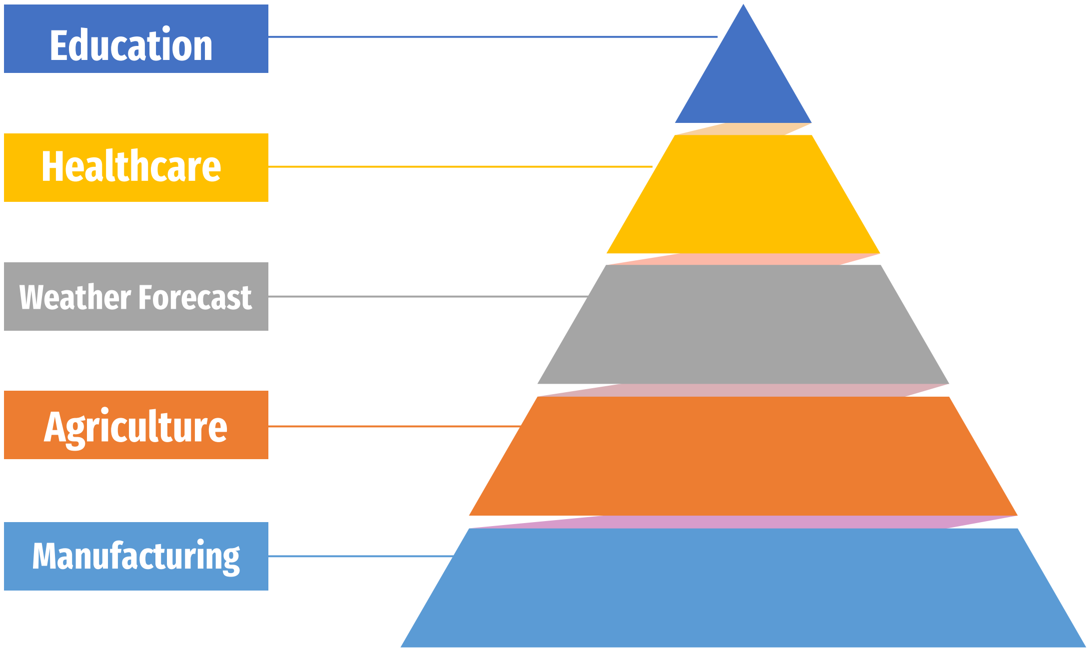
Data Velocity
Real-time
Near Real-time
Periodic
Batch
MB
GB
TB
PB
Data Volume
Table
Database
Photo
Web
Social
Audio
Unstructured
Video
Mobile
Data Variety

Tools used in Big Data Analytics

Tools

MongoDB	MongoDB is used on datasets that change frequently Tools used in big data analytics
Hadoop	**Hadoop helps in storing and analyzing big data Tools used in big data analytics**
Talend	Talend is a tool used for data integration and management Tools used in big data analytics
Cassandra	It is a distributed database that is used for handling chunks of data
Storm	It is an open source real time computational system
Spark	It is used for real time processing and analyzing large amount of data

Big Data application domains



Education

Enhancing Student Results

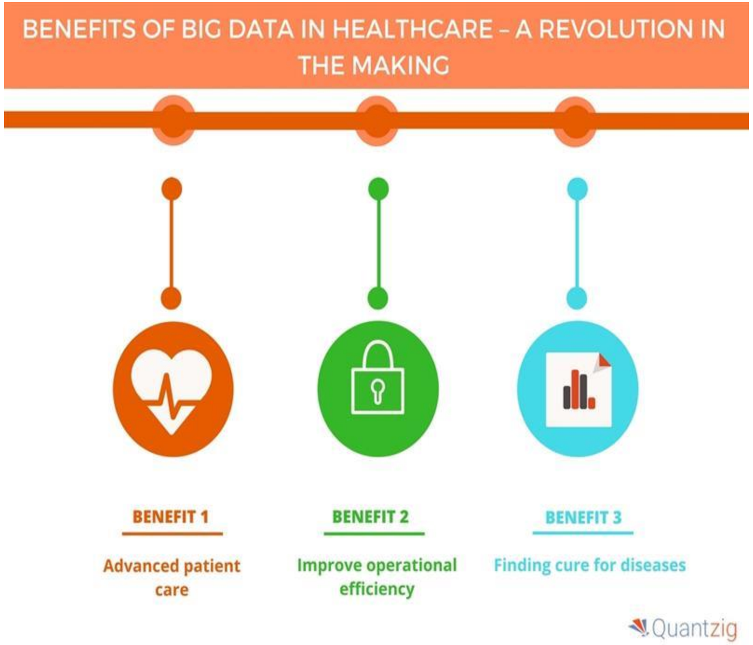
A better Grading System

Gaining Attention

Customized Programs

Reducing The Number of Dropouts

HealthCare



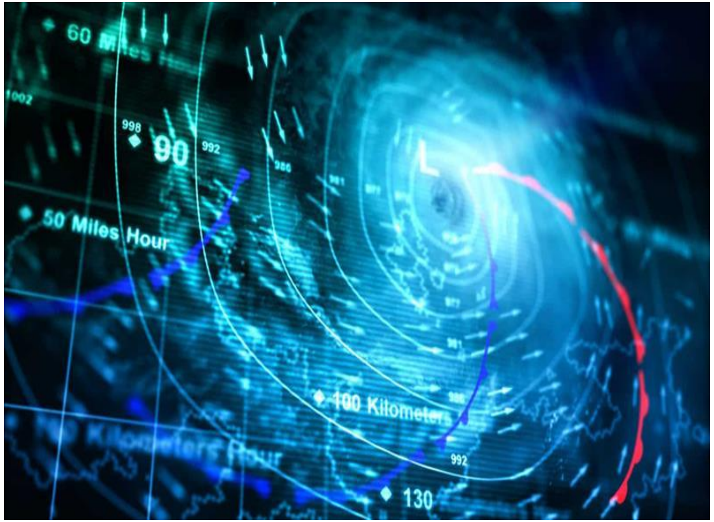
Health Tracking

Improve the care delivery system / machinery

Fraud detection and prevention

Real-time alerts

Weather Forecast



- Estimates of areas where flooding is likely to be most severe
- The strength and direction of tropical storms
- The most likely amount of snow or rain that will fall in a specific area
- The most likely locations of downed power lines
- Estimates of areas where wind speeds are likely to be greatest
- The locations where bridges and roads most likely to be damaged by storms
- The likelihood of flights being cancelled at specific airports

Agriculture

Boosting productivity

Access to plant genome information

Predicting yields

Risk management

Food safety

Savings

Manufacturing

Product quality

Defects tracking

Supply planning

Manufacturing process defect tracking

Testing and simulation of new manufacturing processes

Benefits Big Data Analytics

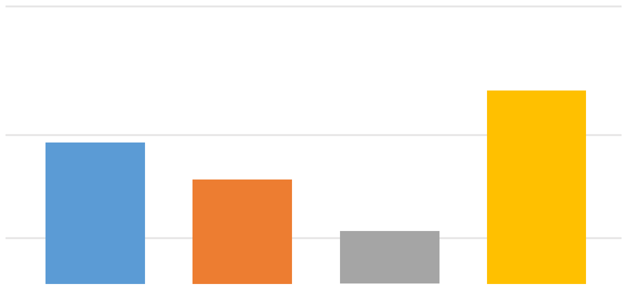
Big data analytics is used for risk management

Big data analytics is used to improve customer experience

Big data analytics is used for product development and innovations

Big data analytics helps in quicker and better decision making in organizations

Google has mastered the domain of big data analytics and it has developed several tools and techniques to capture the data of users which includes their preference, their likes, dislikes, the area of specialization, their requirement etc.



Some Real Time Applications of Big Data

Google Big Data

Google developed several open source tools and techniques that are extensively used in big data ecosystem. With the help of different big data tools and techniques, Google is now capable of exploring millions of websites and **fetch you the right answer or information within milliseconds.**

Google Big Data

The first question that comes to our mind is how can Google perform such complex operations so efficiently?

The answer is Big data analytics.

Big Data tools and techniques to understand our requirements based on several parameters like search history, locations, trends etc.
Google effortlessly displays the complex calculations which are designed to match the user's requirement.

Google always wanted to develop a search engine that has the ability to think like a human and understand the phrase, logic, and goal of any search query. Semantics has helped Google to accomplish this task to look beyond the literal meaning of any phrase of a search query.

Google Has adopted the following techniques using Big Data Analytics

IBM's Weather Forecasting

One example of an application of big data to weather forecasting is IBM’s Deep Thunder. Unlike many weather forecasting systems, which give general information about a broad geographical region, Deep Thunder provides forecasts for extremely specific locations, such as a single airport, so that local authorities can get critically important information in real time.

Challenges with Big Data

- Capture
- Storage (Solution: Cloud Computing)
- Curation (Management of data + Data retention)
- Search
- Analysis
- Transfer
- Visualization
- Privacy violations

4.Distinguish between traditional BI vs BIG DATA

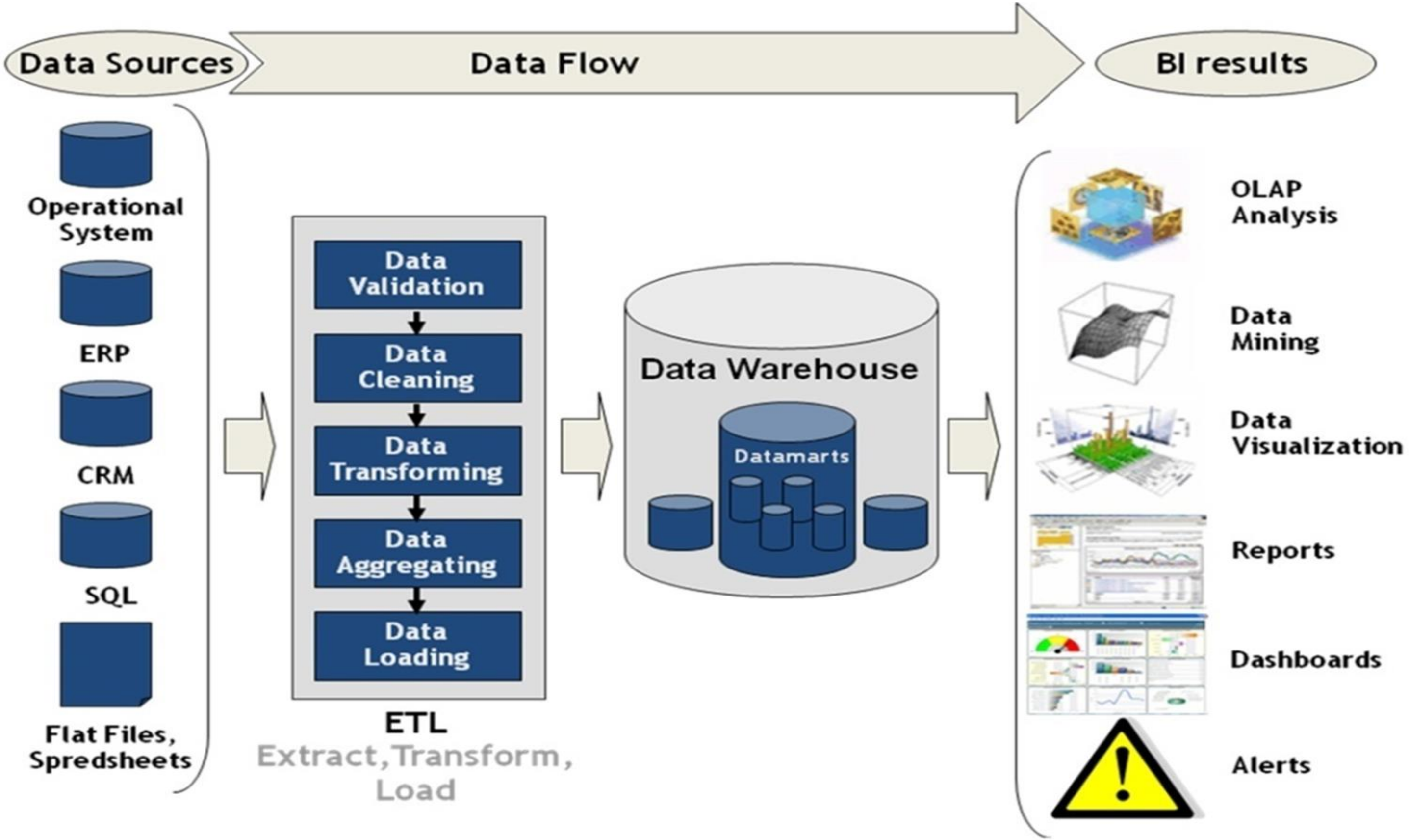
TRADITIONAL BI(BUSINESS INTELLIGENCE)

- All the enterprises data is housed in a central server.
- The database is scaled vertically
- The data is generally analysed in offline mode
- Traditional BI is about structured data

BigData

- The enterprises are housed in the distributed file system.
- The distributed file are scaled in and out
- The data is analysed in online as well as online mode.
- Big data is about variety: Structured ,semi-structured ,unstructured

Typical data warehouse Environment



1. Data Sources – *Where the raw data comes from*

- **Operational System**: The retail store's point-of-sale (POS) system records every transaction.
- **ERP**: Inventory and supply chain data from the Enterprise Resource Planning system.
- **CRM**: Customer loyalty program data, preferences, and feedback.
- **SQL**: Historical sales records stored in databases.
- **Flat Files, Spreadsheets**: Sales reports from different branches in Excel/CSV format.

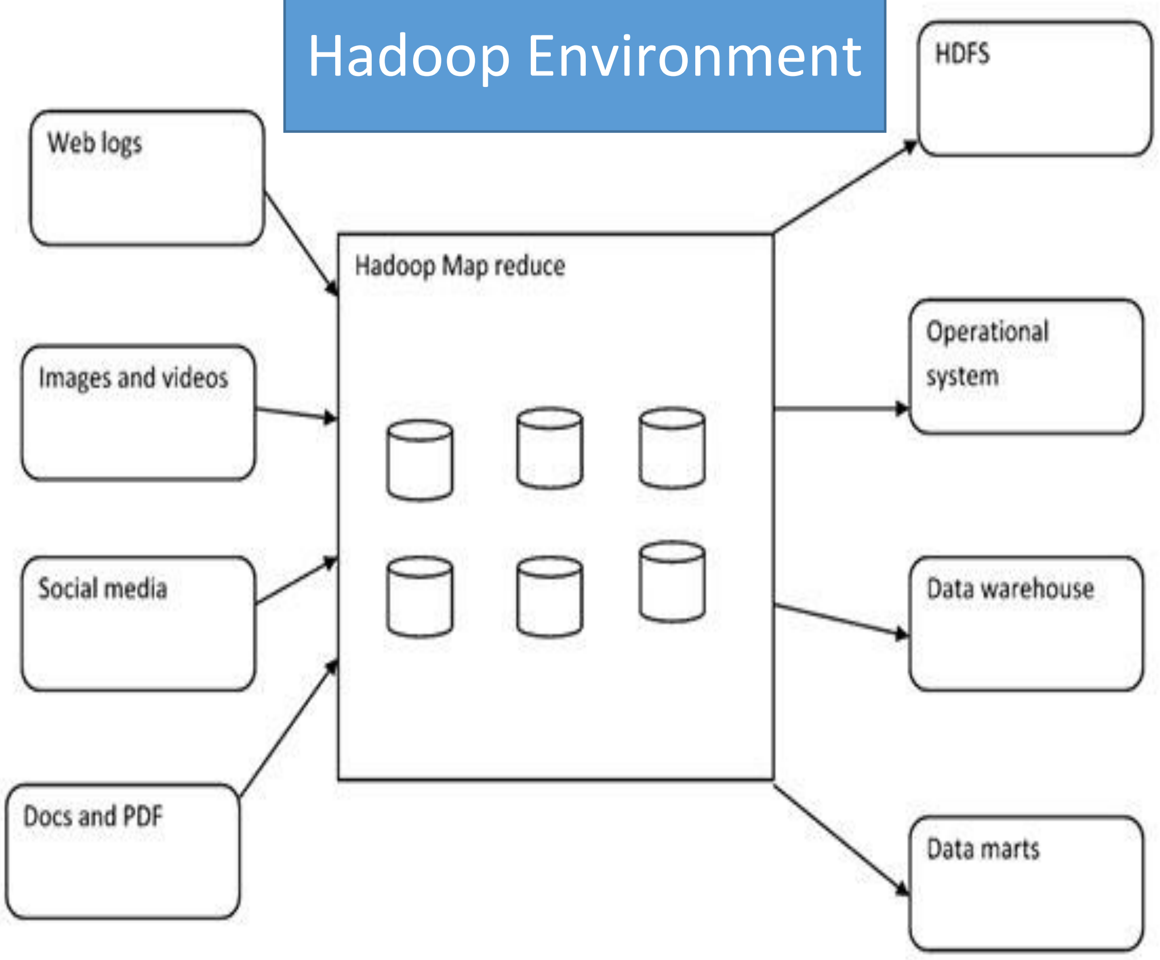
2. Data Flow → ETL Process

- **Data Validation**: Ensure data is complete and correct (e.g., no missing product IDs, correct sales amounts).
- **Data Cleaning**: Remove duplicates, fix formatting issues, correct inconsistent product names.
- **Data Transforming**: Convert all currency values to a single currency, categorize products into standard categories (e.g., "Men's Shoes" instead of "Men Shoes" or "Shoes-Men").
- **Data Aggregating**: Summarize daily sales into weekly or monthly totals for each branch.
- **Data Loading**: Store the processed data into the **Data Warehouse**.

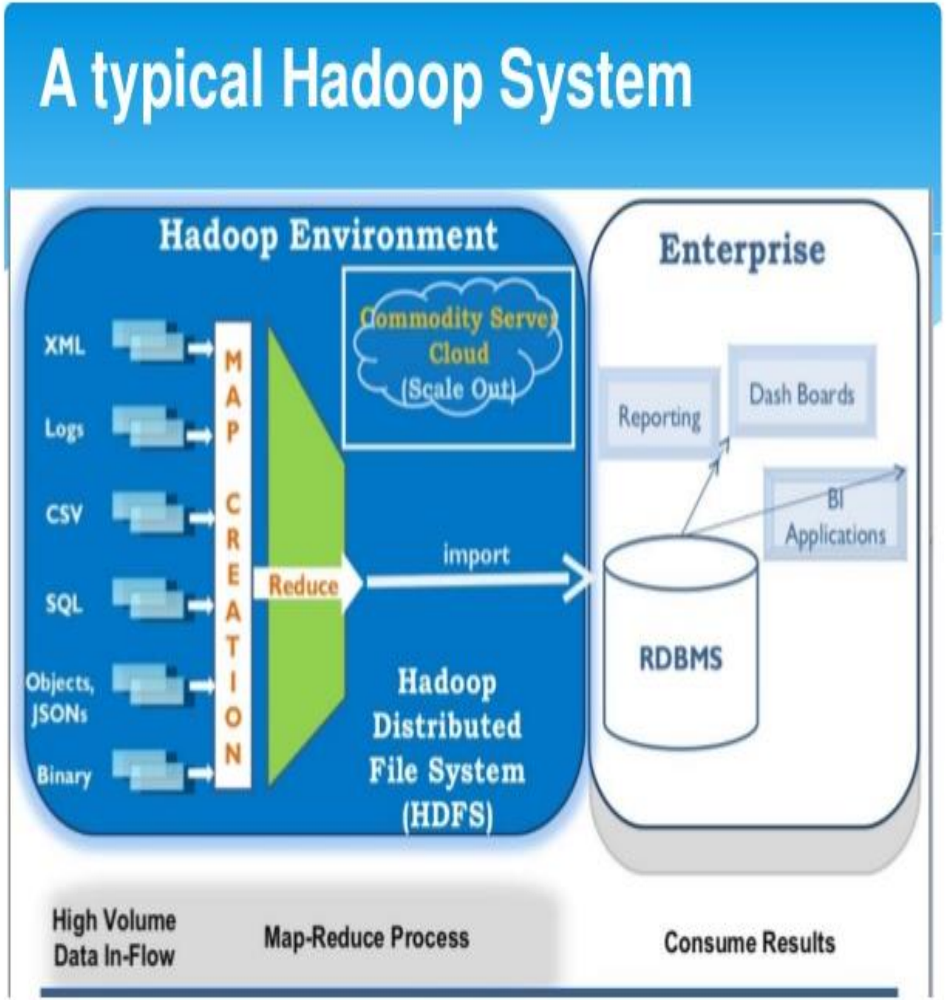
3. Data Warehouse

- Central storage for cleaned and organized data.
- Contains **Datamarts** (smaller, subject-specific databases) such as:
  - Sales datamart
  - Inventory datamart
  - Customer datamart

Hadoop Environment



- **Inputs**: Different types of raw data sources — Web logs, Images & videos, Social media, Docs/PDFs.
- **Processing**: Hadoop MapReduce — the engine that processes all this large, messy, unstructured data.
- **Outputs**: Data stored into systems like:
- **HDFS** (Hadoop Distributed File System)
- **Operational systems**: Updates real time data
- **Data warehouse:** stores processed data for analytics.
- **Data marts:** store summaries for quick marketing queries.



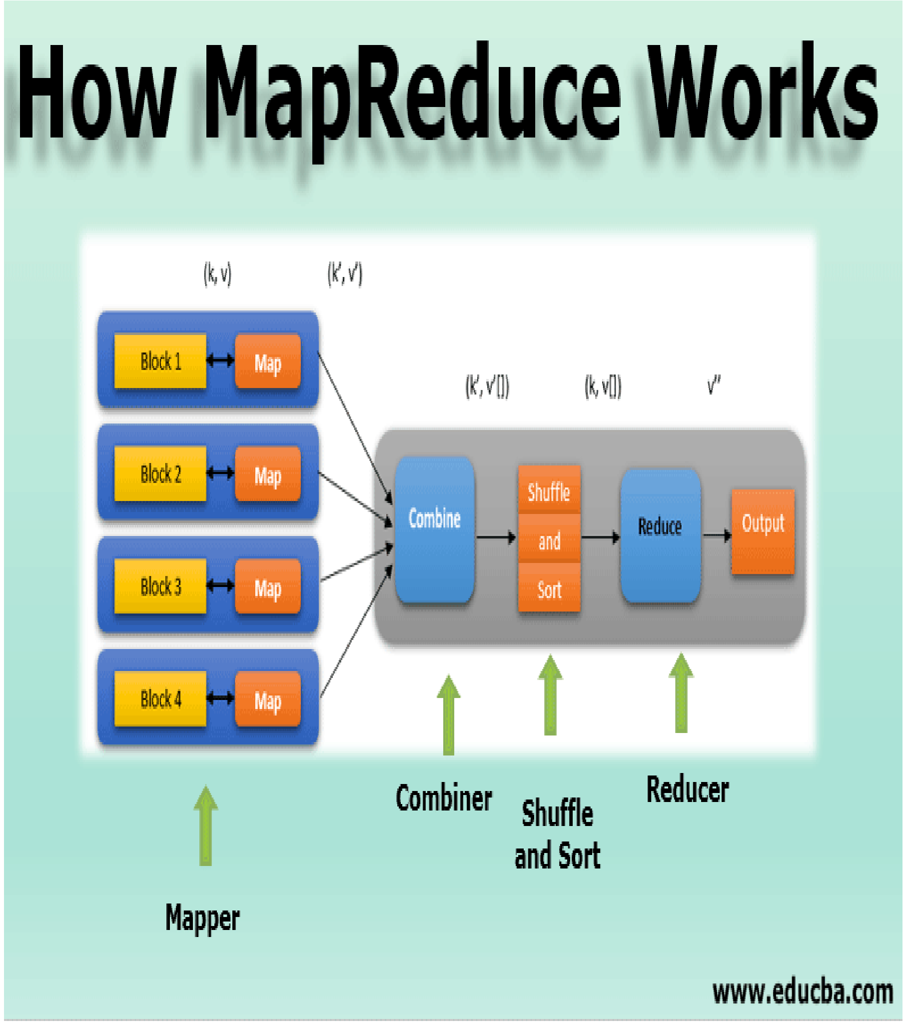
•**High-volume data in-flow**: Data comes in from multiple formats — XML, Logs, CSV, SQL, JSON, Binary.
•**Map Creation**: Breaks large data into smaller chunks to process in parallel.
•**Reduce**: Combines and aggregates the results.
•**HDFS**: Stores the processed data in a distributed way.
•**Enterprise use**: BI applications, dashboards, reporting.

Amazon.com order analysis:

•**Inputs**:

- Logs = Clickstream data (where users clicked).
- CSV = Transaction details.
- SQL = Inventory records.
- JSON = API data from suppliers.
- Binary = Product images.

•**Map Phase**: Each server processes a portion of the data (e.g., count orders per product).
•**Reduce Phase**: Combines all counts to get total sales.
•**HDFS**: Stores large historical datasets.
•**Enterprise**: Dashboards for sales managers, BI tools for trend analysis.



•**Mapper**: Processes each data block individually (e.g., reading a log file line by line, extracting key-value pairs like <productID, 1>).
•**Combiner**: Optional mini-reducer to reduce network load (e.g., summing partial counts locally).
•**Shuffle and Sort**: Groups all data with the same key together (e.g., all sales of the same product).
•**Reducer**: Aggregates the grouped data into final results (e.g., total sales per product).
•**Output**: Stored in files or databases.

Counting hashtags on **Twitter** in real-time:
•**Mapper**: For each tweet, emit <hashtag, 1>.
•**Combiner**: Sum locally (e.g., #AI → 10 from a batch of tweets on one server).
•**Shuffle and Sort**: Send all #AI counts to the same reducer.
•**Reducer**: Final sum for #AI.
•**Output**: “#AI was used 1.2M times today.”

Big data & DW coexistence

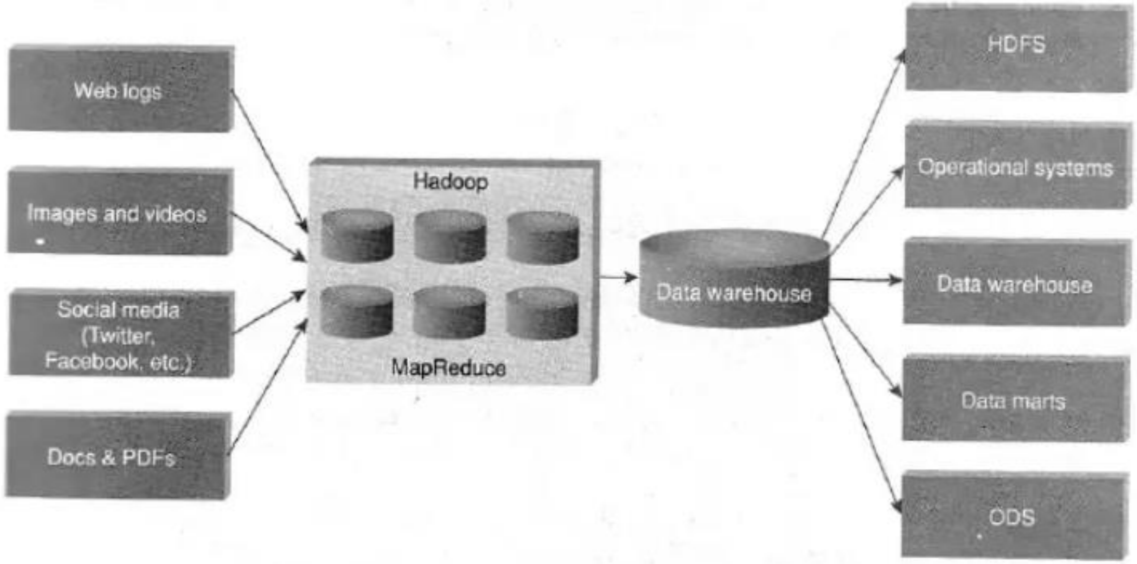


Figure 2.11 Big data and data warehouse coexistence.

1. Big Data Sources

Web logs → Clickstream data from websites (e.g., pages visited, time spent).

Images and videos → Multimedia from surveillance, marketing, product images, etc.

Social media → Posts, likes, shares, comments from platforms like Twitter and Facebook.

Docs & PDFs → Reports, contracts, scanned documents

2. Hadoop & MapReduce (Processing Layer)

- **Hadoop** stores massive amounts of raw data using **HDFS (Hadoop Distributed File System)**.
- **MapReduce** processes this data in parallel across many machines — perfect for analytics on huge datasets.
- At this stage, big data is processed, filtered, and transformed into structured or semi-structured formats

3. Data Warehouse Integration

- The **processed results** from Hadoop are loaded into a **Data Warehouse**.
- The Data Warehouse is optimized for structured data and fast querying.
- This allows big data insights to be combined with traditional business data for **comprehensive analysis**.

Data Warehouse Outputs

HDFS → Some processed data may still be stored back into Hadoop for long-term big data storage.

- **Operational systems** → Transaction systems (ERP, CRM) may use processed insights for real-time decisions.
- **Data warehouse** → The main repository for structured, historical business data.
- **Data marts** → Smaller, subject-specific slices of the data warehouse (e.g., sales, marketing).
- **ODS (Operational Data Store)** → For near real-time operational reporting.

Thank You